Producer Behavior

Chapters 6

1 The Basics of Production

> **Final Good:** A good that is bought by a consumer.

> **Intermediate Good:** A good that is used to produce another good.

> **Production:** The process by which a person, company, government, or non-profit agency uses inputs to create a good or service for which others are willing to pay.

1.1 Simplifying Assumptions about Firm's Production Behavior

1. The firm produces a single good.
2. The firm has already chosen which product to produce. *This decision is studied in an IO class.*
3. For whatever quantity it makes, the firm's goal is to minimize the cost of producing it. *The goal is not to minimize overall cost of production, because doing so would result in zero output.*
4. The firm uses only two inputs to make its product: capital and labor.
5. In the short run, a firm can choose to employ as much or as little labor as it wants, but it cannot rapidly change how much capital it uses. In the long run, the firm can freely choose the amounts of both labor and the capital it employs.

   > **Short-Run:** In production economics, the period of time during which one or more inputs into production cannot be changed.

   > **Fixed Inputs:** Inputs that cannot be changed in the short-run.

   > **Variable Inputs:** Inputs that can be changed in the short-run.

   > **Long-Run:** In production economics, the amount of time necessary for all inputs into production to be fully adjustable.

6. The more inputs the firm uses, the more output it makes.
7. A firm's production exhibits diminishing marginal returns to labor and capital.
8. The firm can buy as many capital or labor inputs as it wants at fixed market prices.
9. If there is a well-function capital market (e.g. banks and investors), the firm does not have a budget constraint. *As long as the firm can make profits, it will be able to obtain the resources necessary to acquire the capital and labor it needs to produce. If a firm doesn't have the cash necessary to finance its input expenditures, it can raise funds by borrowing (or issuing stock).*

POLL EVERYWHERE: A basic assumption of production is that the firm:

1.2 Production Functions

> **Production Function:** A mathematical relationship that describes how much output can be made from different combinations of inputs.

$$Q = f(K, L)$$

$$Q = 10K + 5L$$
$$Q = K^{0.5}L^{0.5}$$

The later example is known as the Cobb-Douglas production function. It usually describes actual production data reasonably well and it has handy mathematical properties.

2 Production in the Short-Run

> **Marginal Product:** The additional output that a firm can produce by using an additional unit of an input (holding use of the other input constant).

$$MP_L = \frac{\Delta Q}{\Delta L}$$

> **Diminshing Marginal Product:** The reduction in the incremental output obtained from adding more and more labor.

> **Average Product:** total quantity of output divided by the number of units of input used to produce it.

$$AP_L = \frac{Q}{L}$$

Suppose we had the following production function:

$$Q = K^{0.5}L^{0.5}$$

and in the short-run, the capital stock was fixed at 4 units. This means our short-run production function will be:

$$Q = 4^{0.5}L^{0.5} = 2L^{0.5}$$

Capital (K)	Labor (L)	Output (Q)	Marginal Product of Labor (MP_L)	Average Product of Labor (AP_L)
4	0	0	—	—
4	1	2	2	2
4	2	2.83	0.83	1.42
4	3	3.46	0.63	1.15
4	4	4	0.54	1
4	5	4.47	0.47	0.89

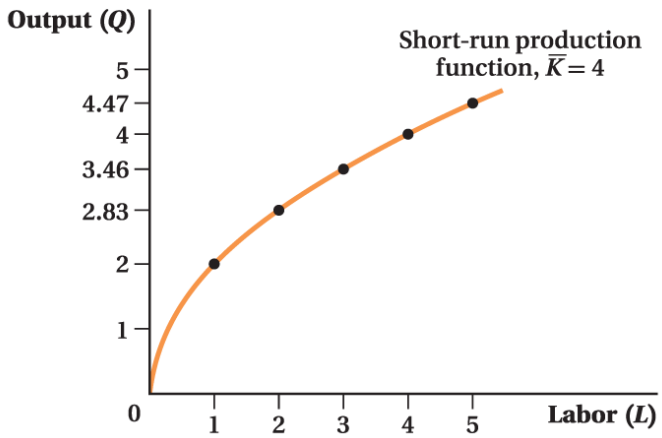

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Figure 1: This figure graphs the firm's continuous short-run production function using the values from Table 6.1. The production function's positive slope means that an increase in labor increases output. As the firm hires more labor, however, output increases at a decreasing rate and the slope flattens.

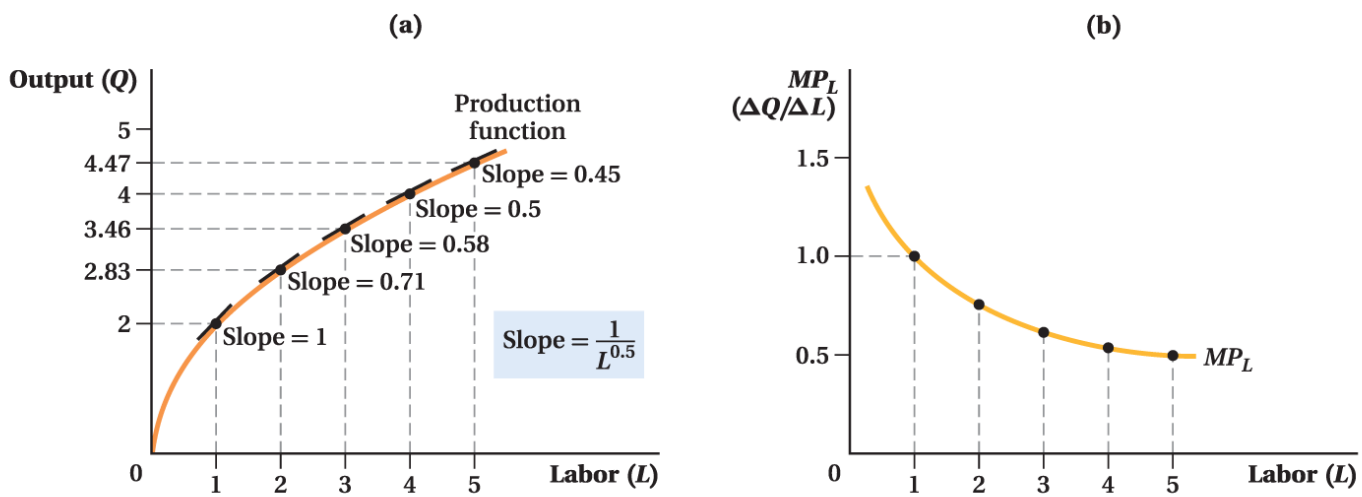

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Figure 2: (a) The marginal product of labor is the slope of the production function. As the quantity of labor increases, the marginal product decreases, from $MP_L = 1$ when $L = 1$ to $MP_L = 0.45$ when $L = 5$, and the slope flattens. (b) Using the production function in panel a, we can derive the marginal product of labor curve. The downward slope of the curve shows the diminishing marginal returns to labor.

3 Production in the Long-Run

In the long run, firms can change not only their labor inputs but also their capital. This difference gives them two important benefits:

1. In the long run, a firm might be able to lessen the sting of diminishing marginal products.
2. In the long run is that producers often have some ability to substitute capital for labor or vice versa.

The Long-Run production function is:

$$Q = f(K, L) = K^{0.5}L^{0.5}$$

4 The Firm's Cost Minimization Problem

> **Cost Minimization:** A firm's goal of producing a specific quantity of output at minimum cost.

4.1 Isoquants

> **Isoquant:** A curve representing all the combinations of inputs that allow a firm to make a particular quantity of output.

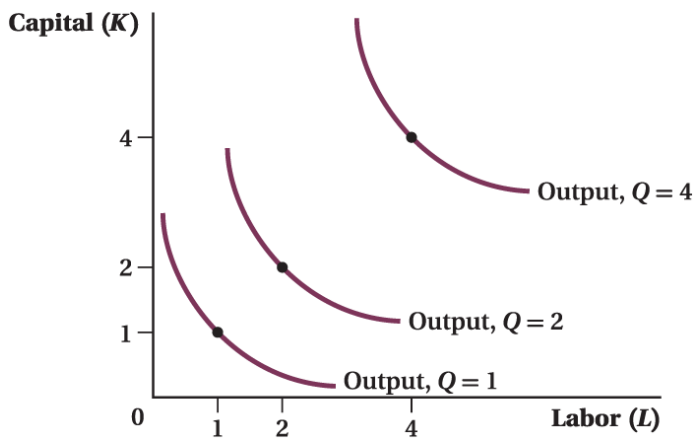

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Figure 3: Each isoquant shows the possible combinations of labor (L) and capital (K) that produce the output (Q) levels 1, 2, and 4 units.

4.1.1 The Marginal Rate of Technical Substitution

Marginal Rate of Technical Substitution ($MRTS_{XY}$): The rate at which the firm can trade input X for input Y, holding output constant.

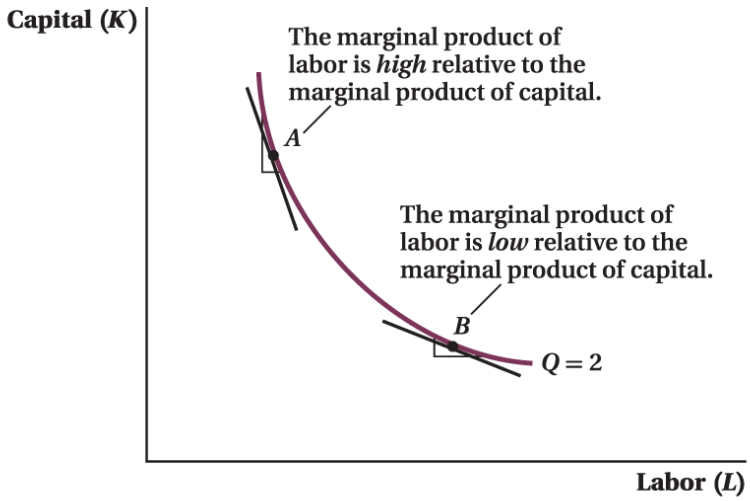

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Figure 4: The negative slope of the isoquant is the marginal rate of technical substitution of labor L for capital K. At point A, the marginal product of labor is high relative to the marginal product of capital, and a relatively small decrease in labor would require a large quantity of capital to hold output constant. At point B, the marginal product of labor is low relative to the marginal product of capital, and a relatively small decrease in capital would require a large quantity of labor to hold output constant.

$$\begin{aligned}
\Delta Q = MP_L \times \Delta L + MP_K \times \Delta K &= 0 \\
MP_K \times \Delta K &= -MP_L \times \Delta L \\
\frac{\Delta K}{\Delta L} &= -\frac{MP_L}{MP_K} \\
MRTS_{LK} = -\frac{\Delta K}{\Delta L} &= \frac{MP_L}{MP_K}
\end{aligned}$$

Therefore, the $MRTS_{LK}$ at any point on an isoquant tells you the relative marginal products of capital and labor at that point.

4.1.2 Substitutability

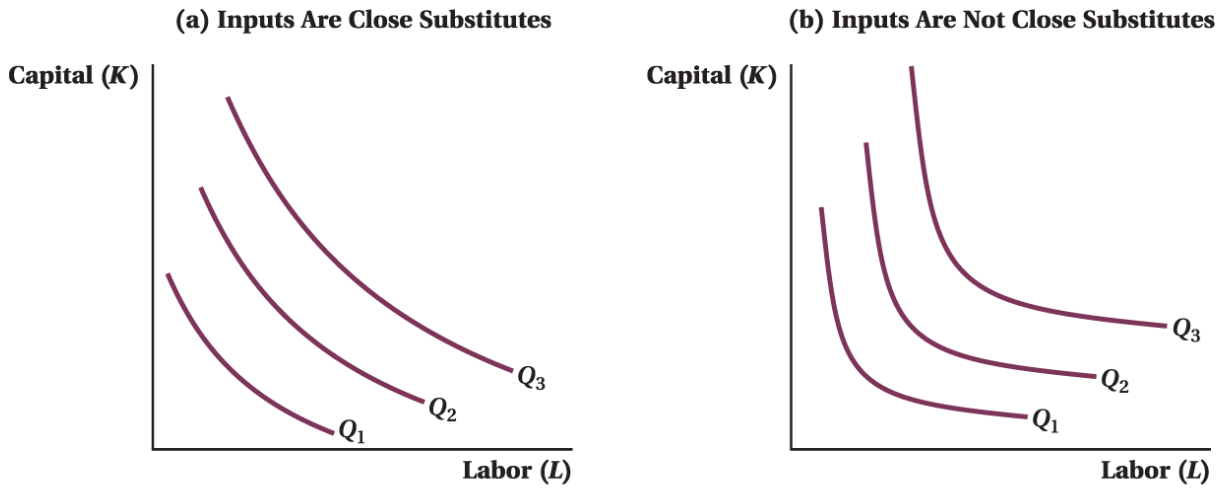

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Figure 5: (a) Relatively straight isoquants indicate that $MRTS_{LK}$ does not vary much along the curve. Therefore, labor and capital are close substitutes for each other. (b) Relatively curved isoquants indicate that $MRTS_{LK}$ varies greatly along the curve. Therefore, labor and capital are not close substitutes for each other.

4.1.3 Perfect Substitutes and Perfect Complements in Production

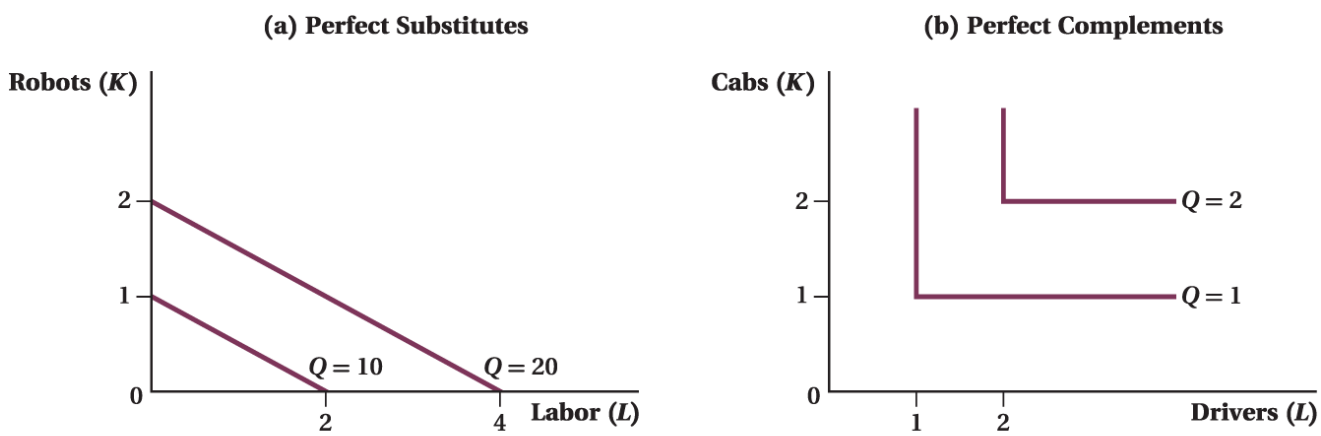

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Figure 6: (a) Robots K and labor L are perfect substitutes. The isoquants are straight lines, and the $MRTS_{LK}$ does not change along the isoquant. In this case, 2 humans can substitute for 1 robot. (b) Cabs K and drivers L are perfect complements. The isoquants are L-shaped, and the optimal quantity (K, L) for each output Q is the corner of the isoquant. In this case, 1 cab with 1 driver produces $Q = 1$, while 2 cabs with 2 drives produce $Q = 2$.

4.2 Isocost Lines

> **Isocost Line:** A curve that shows all the input combinations that yield the same cost.

$$C = RK + WL$$

Two things about Isocost Lines:

- First, isocost lines for higher total expenditure levels are further from the origin. This reflects the fact that, as a firm uses more inputs, total expenditure on those inputs increases.
- Second, the isocost lines are parallel (assumption 8). They all have the same slope, regardless of what total cost level they represent.

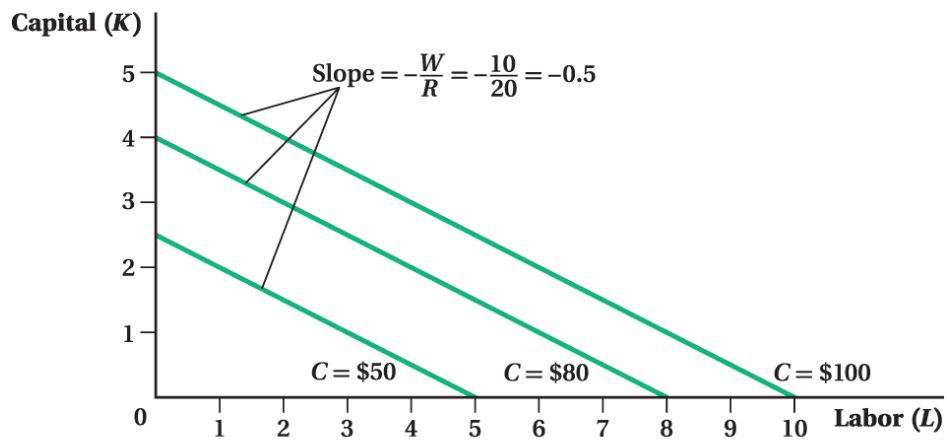

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Figure 7: Each isocost line shows all bundles of inputs that have the same cost to the firm. As you move farther from the origin, the isocost lines represent higher total expenditures, from cost $C = \$50$ to $C = \$80$. For wage $W = \$10$ and rental rate $R = \$20$, the slope of the isocost lines is $-W/R$ or $-1/2$. Therefore, for every 1-unit increase in K, the firm has to give up 2 units of L in order to keep cost constant.

$$\begin{aligned} C &= RK + WL \\ RK &= C - WL \\ K &= \frac{C}{R} - \frac{W}{R}L \end{aligned}$$

As is so often the case in economics, the slope tells us about tradeoffs at the margin. Here, the slope reflects the cost consequences of substituting one input for another. It indicates how much more of one input a firm could hire, without increasing overall expenditure on inputs, if the firm used less of the other input.

POLL EVERYWHERE: Suppose the wage rate is $25 per hour and the rent on capital is $50 per hour. The equation for the isocost line is given by:

4.3 Isocost Lines and Input Price Changes

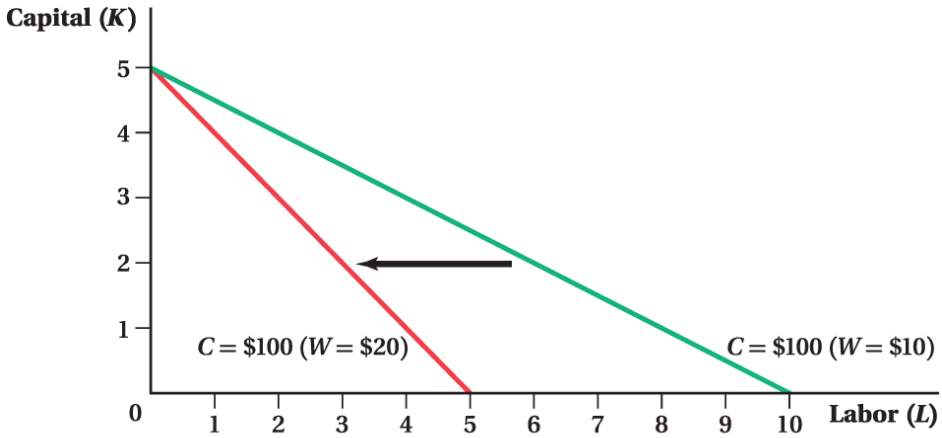

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Figure 8: Increase in wage from $W = \$10$ to $W = \$20$.

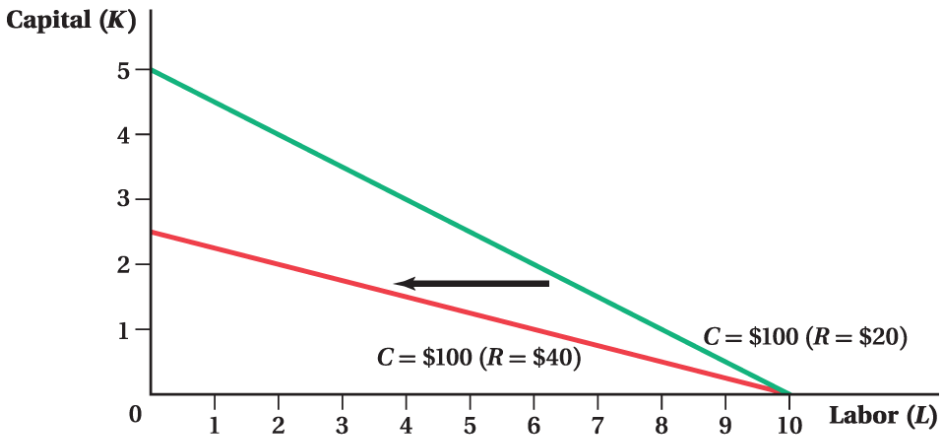

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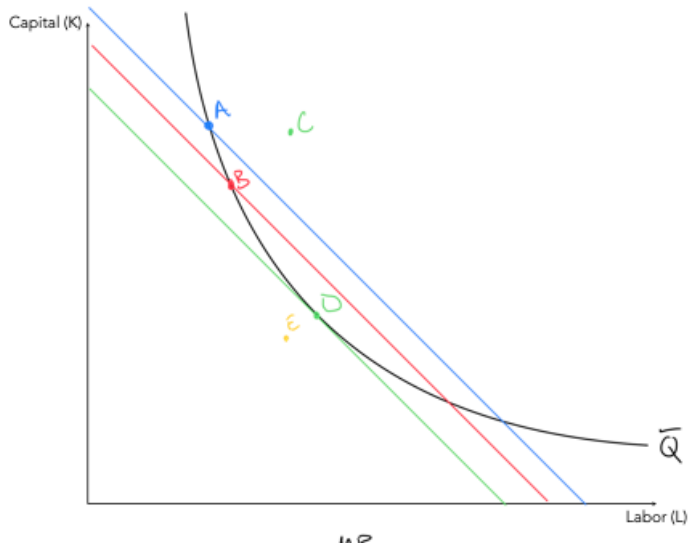
Figure 9: Increase in the rental rate from $R = \$20$ to $R = \$40$.

4.4 Identifying Minimum Cost: Combining Isoquants and Isocost Lines

- The firm's objective is to minimize costs subject to the constraint that it has to produce a particular quantity of output, $\bar{Q}$.
- This means that the firm wants the isocost curve as close to the origin as possible.



- Input bundle A will produce the desired amount of output, but because portions of the Isoquant are below A's Isocost Line, there are cheaper ways to produce $\bar{Q}$.
- Input bundle B will produce the desired amount of output, but because portions of the Isoquant are below B's Isocost Line, there are cheaper ways to produce $\bar{Q}$.
- Input bundle C produces more output than is required, therefore the firm could always lower costs by producing less output.
- Input bundle D produces the desired amount of output at the cheapest cost. Why?
  - Any other combination of inputs that produces $\bar{Q}$ lies above D's Isocost Line and are therefore more expensive.
  - Any combination of inputs that is cheaper than bundle D lies below the $\bar{Q}$ Isoquant and therefore does not produce the desired amount of output.

Another important feature of the tangency result is that the isocost line and isoquant have the same slope at a tangency point (that's the definition of a tangency point).

$$\frac{W}{R} = \frac{MP_L}{MP_K}$$
$$\frac{MP_K}{R} = \frac{MP_L}{W}$$

One way to interpret these ratios is that they measure the marginal product per dollar spent on each input, or the input's "bang for the buck." Alternatively, we can think of each of these ratios as the firm's marginal benefit-to-cost ratio of hiring an input.

Cost minimization implies that each input's benefit-to-cost ratio is equal because only then is the firm unable to reduce the cost of producing its current quantity by changing its input levels.

For example, if the firm produced an input bundle where this wasn't true, if, say, $\frac{MP_K}{R} > \frac{MP_L}{W}$, then the firm's benefit-to-cost ratio for capital is higher than for labor. This would mean that the firm could replace some of its labor with capital while keeping its output quantity the same but reducing its total cost.

4.5 Input Price Changes

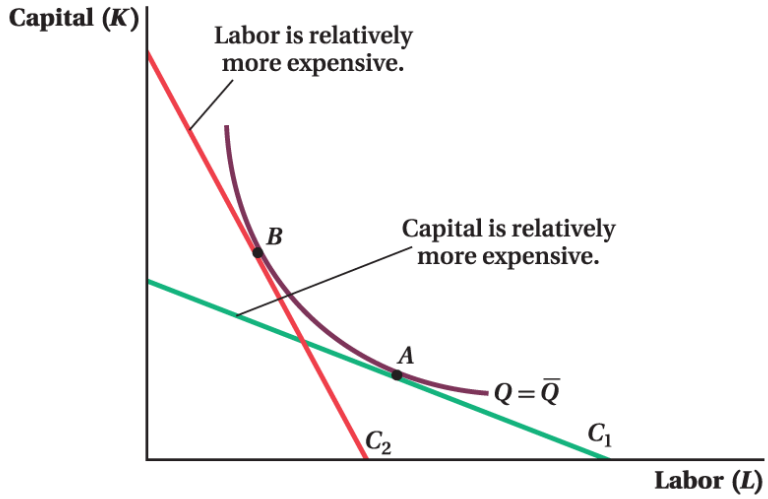


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Figure 10: When labor becomes relatively more expensive, the isocost line shifts from C_1 to C_2. With the steeper isocost line, the cost-minimizing input choice shifts from point A, with a high ratio of labor to capital, to point B, with a low ratio of labor to capital.

5 Returns to Scale

> **Returns to Scale:** A change in the amount of output in response to a proportional increase in all the inputs.

> **Constant Returns to Scale:** A production function for which changing the amount of capital and labor by some multiple changes the quantity of output by exactly the same multiple.

> **Increasing Returns to Scale:** A production function for which changing all inputs by the same proportion changes output more than proportionately.

> **Decreasing Returns to Scale:** A production function for which adjusting all inputs by the same multiple changes output by less than that multiple.

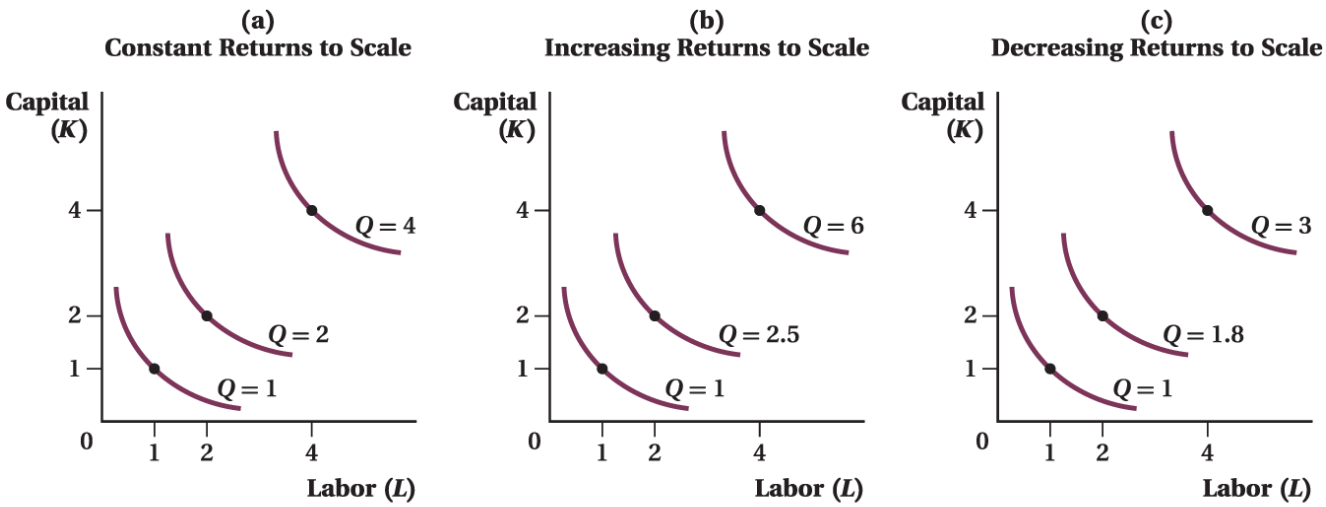

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Figure 11

5.1 Factors Affecting Returns to Scale

Constant returns to scale are natural to think of (building a second identical factory).

Increasing returns to scale come about through fixed costs or learning-by-doing.

> **Fixed Cost:** An input cost that does not vary with the amount of output, even if output is zero.

> **Learning-by-doing:** The process by which a firm becomes more efficient at production as it produces more output.

Decreasing returns to scale are possible but should be unlikely in the long run for the same reason that constant returns to scale are natural. If inputs are measured properly and the firm has sufficient time to adjust all its inputs, the firm can just replicate its current production operation, allowing it to increase output by the same factor as inputs.

POLL EVERYWHERE: What describes increasing returns to scale?

6 Technological Change

> **Total Factor Productivity Growth (technological change):** An improvement in technology that changes the firm's production function such that it gets more output from the same amount of inputs.

$$Q = Af(K, L)$$

How does an increase in A effect the firm's cost minimization problem?

- First, the firm's isocost lines don't change because the prices of the inputs remain fixed.
- Second, since A is part of the production function, it will affect the isoquants.

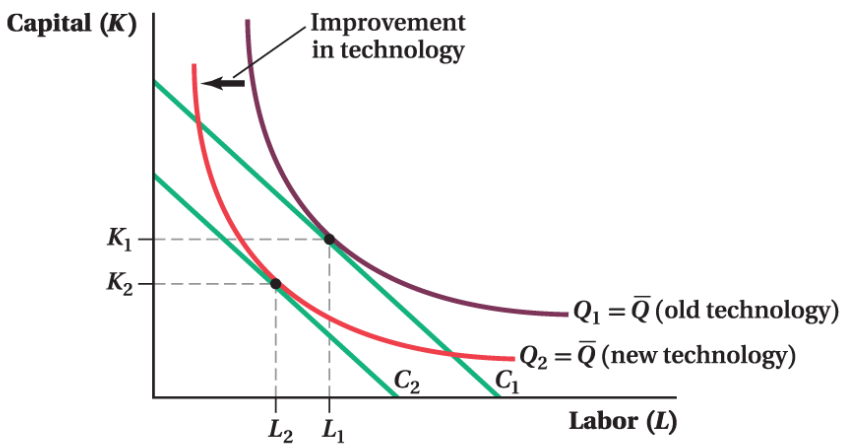

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Figure 12

7 The Firm's Expansion Path and Total Cost Curve

> **Expansion Path:** A curve that illustrates how the optimal mix of inputs varies with total output.

> **Total Cost Curve:** A curve that shows a firm's cost of producing particular quantities.

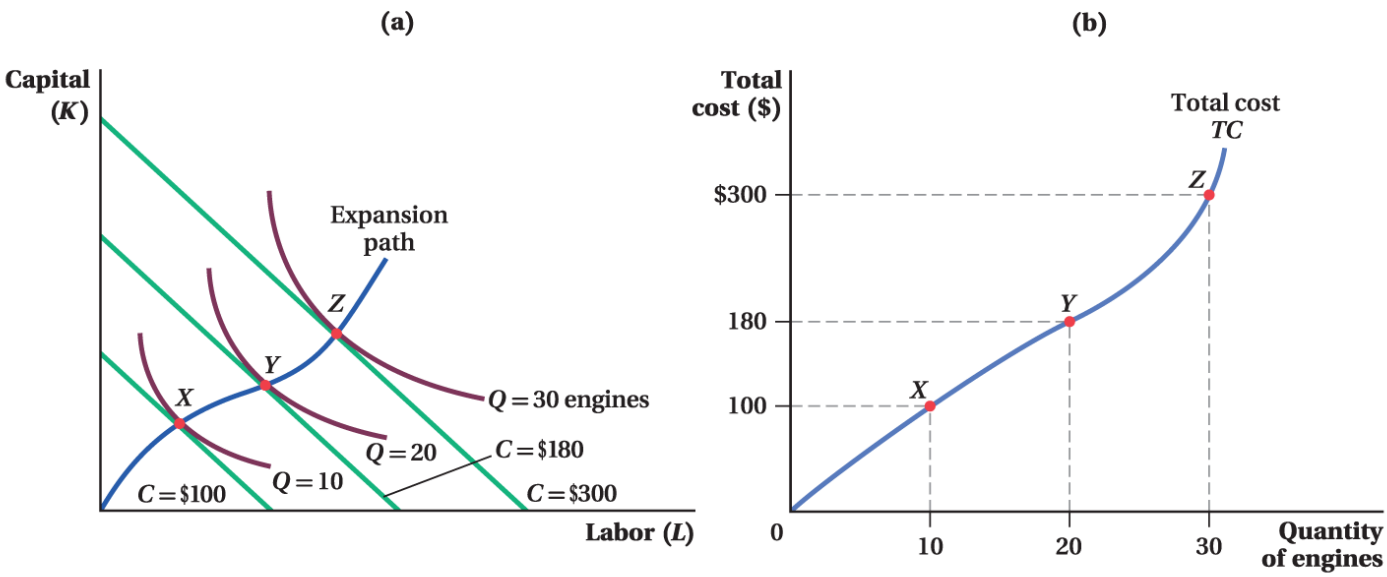

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Figure 13

8 Appendix: The Calculus of Cost Minimization

- Like in chapter 4, we are going to assume a Cobb-Douglas functional form:

$$Q = AK^{\alpha}L^{1-\alpha}$$

- Earlier in this chapter, we saw that the cost minimizing point for the firm occurs when:

$$MRTS_{LK} = \frac{W}{R}$$

- And that:

$$MRTS_{LK} = \frac{MP_L}{MP_K}$$

- Therefore,

$$\frac{MP_L}{MP_K} = \frac{W}{R}$$

- Using this equation we can find the optimal amount of capital and labor to hire. First, we need to solve for the marginal products of capital and labor:

$$MP_L = (1-\alpha)AK^{\alpha}L^{-\alpha}$$
$$MP_K = \alpha AK^{\alpha-1}L^{1-\alpha}$$

- Next, we can solve for the marginal rate of technical substitution:

$$MRTS_{LK} = \frac{(1-\alpha)AK^{\alpha}L^{-\alpha}}{\alpha AK^{\alpha-1}L^{1-\alpha}} = \frac{1-\alpha}{\alpha}\frac{K}{L}$$

- Substituting this into our optimality condition yields:

$$\frac{1-\alpha}{\alpha}\frac{K}{L} = \frac{W}{R}$$
$$K = \left[\frac{\alpha}{1-\alpha}\frac{W}{R}\right] L$$

- Note that the equation above is the firm's expansion path.
- Now the firm's problem is different that the consumer's. The consumer is trying to maximize their utility subject to a specific amount of income. The firm is trying to minimize its costs for a specific amount of output it needs to produce.
- From the equation above, we know how much capital the firm will hire based on the number of workers it has, we now just need to figure out how many workers the firm needs to hire.
- Suppose the firm has a production target $\bar{Q}$. We can solve for the optimal amount of hire by substituting the relationship between capital and labor into the firm's production function:

$$\bar{Q} = AK^{\alpha}L^{1-\alpha}$$
$$\bar{Q} = A\left[\frac{\alpha}{1-\alpha}\frac{W}{R}L\right]^{\alpha} L^{1-\alpha}$$
$$\bar{Q} = A\left[\frac{\alpha}{1-\alpha}\frac{W}{R}\right]^{\alpha} L$$
$$L = \left[\frac{1-\alpha}{\alpha}\frac{R}{W}\right]^{\alpha}\frac{\bar{Q}}{A}$$

- This implies that the optimal amount of capital to hire is:

$$K = \left[\frac{\alpha}{1-\alpha}\frac{W}{R}\right] L$$
$$K = \left[\frac{\alpha}{1-\alpha}\frac{W}{R}\right]\left[\frac{1-\alpha}{\alpha}\frac{R}{W}\right]^{\alpha}\frac{\bar{Q}}{A}$$
$$K = \left[\frac{\alpha}{1-\alpha}\frac{W}{R}\right]\left[\frac{\alpha}{1-\alpha}\frac{W}{R}\right]^{-\alpha}\frac{\bar{Q}}{A}$$
$$K = \left[\frac{\alpha}{1-\alpha}\frac{W}{R}\right]^{1-\alpha}\frac{\bar{Q}}{A}$$

8.1 Example

- Suppose we have the following problem:
  - Production Function: $Q = 20K^{0.2}L^{0.8}$
  - R=$15
  - W=$10
  - Desired production: 40,000 units of output.
- Start by finding the marginal products of labor and capital

$$MP_L = 16K^{0.2}L^{-0.2}$$
$$MP_K = 4K^{-0.8}L^{0.8}$$

- Set the marginal rate of technical substitution equal to the ratio of the prices:

$$\begin{aligned}\frac{MP_L}{MP_K} = \frac{16K^{0.2}L^{-0.2}}{4K^{-0.8}L^{0.8}} &= \frac{10}{15} \\ \frac{4K}{L} &= \frac{10}{15} \\ 60K &= 10L \\ K &= \frac{1}{6}L\end{aligned}$$

- Plug this into the production function to find the optimal amount of labor:

$$\begin{aligned}40,000 &= 20\left[\frac{1}{6}L\right]^{0.2} L^{0.8} \\ 2,000 &= \left[\frac{1}{6}\right]^{0.2} L \\ 2,000 &= 0.6988L \\ L &= 2,861.94 \\ K &= 476.99\end{aligned}$$